Status Report Number 2

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1 September 2026

1 Architecture and intervention ladder

Figure 1 locates the activation sites and organizes the progressive experimental program.

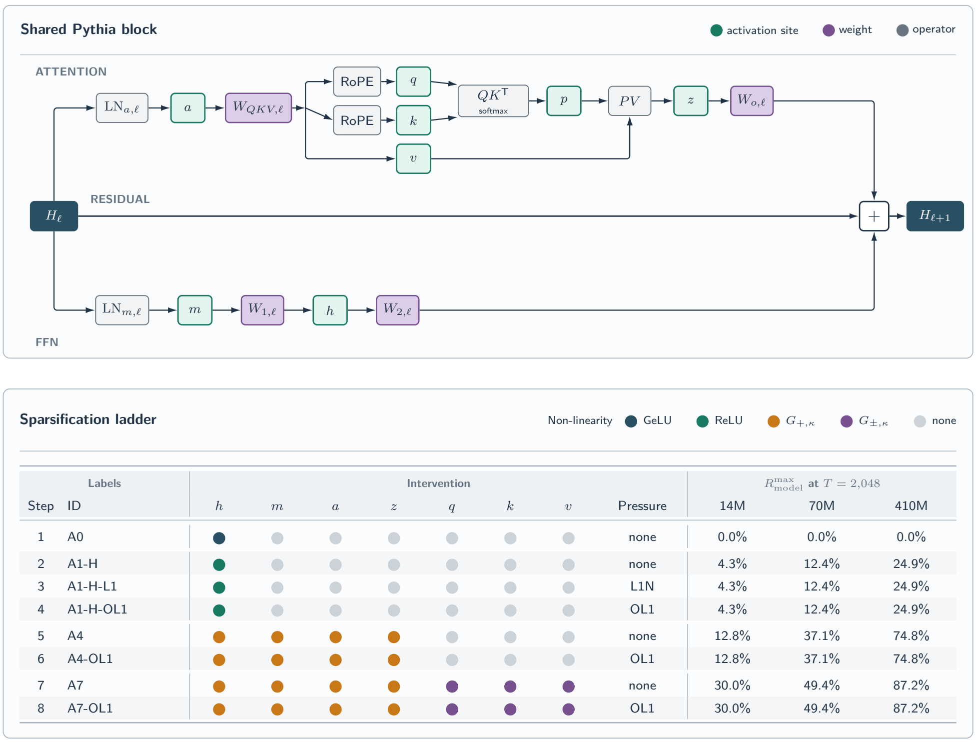

Figure 1: Shared Pythia block and sparsification intervention ladder. A4 and A7 denote operational topologies A4-Z and A7-Z-POST.

2 Experiment control

2.1 Scope dashboard

Pilot. Local execution with 581 optimizer boundaries, global batch 64, and 76,152,832 training input tokens per condition.

Paper-scale full pass. RunPod A100 execution at 14M and H200 execution at 70M, with 712 optimizer boundaries, global batch 1,024, and 1,493,172,224 training input tokens per condition. Conditions run concurrently across independent GPUs.

Table 1: Experiment scope at the report cutoff.

Step	Conditions/model	Pythia-14M	70M	410M
A0	1	Full pass complete: Run 004	1/1 + TEAL: Run 018	0/1
A1-H	1	Full pass complete: Run 004	1/1 + TEAL: Run 018	0/1
A1-H-L1	4	4/4 full pass: Run 004	0/4	0/4
A1-H-OL1	4	4/4 full pass: Run 009	0/4	0/4
A4	5	5/5 full pass: Run 011	0/5	0/5
A4-OL1	5	5/5 full pass: Run 015	5/5 full pass: Run 018	0/5
A7	5	5/5 full pass: Run 013	0/5	0/5
A7-OL1	5	5/5 full pass: Run 014	5/5 full pass: Run 018	0/5

2.2 Paper-scale contract and parameter set

The scale plan now adopts the size-specific learning rates in the standard Pythia recipe: peak/minimum $10^{-3}/10^{-4}$ for 14M and 70M, and $3 \times 10^{-4}/3 \times 10^{-5}$ for 410M. The study applies each activation intervention within this established recipe.

Table 2: Shared paper-scale execution contract.

Field	Specification
Model construction	Random initialization from pinned Pythia architecture; model seed 1234
Data	Pinned MiniPile cache; data-order seed 1234; sequence length 2,048
Training horizon	712 optimizer boundaries; 1,493,172,224 input tokens per condition
Global batch	1,024 sequences; hardware-specific microbatch and accumulation preserve the product
Optimizer	AdamW, betas $(0.9, 0.95)$, epsilon 10^{-8}, weight decay 0.1, global clipping 1.0
Schedule	1% linear warmup; cosine decay to 10% of the size-specific peak learning rate
Precision	Dynamic FP16 autocast; FP32 parameters and optimizer state
Validation	500 documents; 338 complete blocks; 692,224 input tokens; 1,444-token tail recorded
Retention	Activation counts and moments, weight norms, logical counts, pressure-gradient metrics, final checkpoint

Table 3: Aggregated intervention parameter set. Threshold equality survives.

Family	Gate and pressure	Parameter grid
A0	Stock GeLU at `h`	Singleton
A1-H	ReLU at `h`	Singleton
A1-H-L1	ReLU; naive L1 at `h`	$\lambda \in \{0.05, 0.1, 0.5, 1.0\}$
A1-H-OL1	ReLU; OL1 at `h`	Same λ grid; budget 1.0
A4	One-sided threshold at `a,m,h,z`	$\kappa \in \{0, 0.01, 0.05, 0.1, 0.5\}$
A4-OL1	A4 gates; OL1 at four sites	Same κ grid; $\lambda = 1$; budget 1.0
A7	A4 plus symmetric `q_post,k_post,v` gates	Same κ grid
A7-OL1	A7 gates; OL1 at seven sites	Same κ grid; $\lambda = 1$; budget 1.0

3 RunPod compute

Table 4 reports elapsed wall-clock time for each accepted condition's scientific process. A RunPod REST v2 refresh at 2026-09-01 20:07 UTC reconciled every listed cost to the exact Pod IDs. Current Pod spend across the seven main report runs is \$237.61; the retained 100 GB network volume remains separate at \$0.01/hour (about \$7/month).

Table 4: RunPod execution record. Costs include GPU and temporary Pod-disk charges for every identified preflight, accepted worker, and infrastructure retry.

Run	**Compute**	**Wall-clock**	**Cost**
004	5× A100 80 GB; six conditions	63–108 min	\$21.52
009	4× A100 80 GB; four conditions	86–105 min	\$14.58
011	5× A100 80 GB; five conditions	59–104 min	\$16.11
015	5× A100 80 GB; five conditions	94–101 min	\$18.45
013	5× A100 80 GB; five conditions	82–89 min	\$15.80
014	5× A100 80 GB; five conditions	74–115 min	\$19.42
018	H200 141 GB; 12 conditions; 15 total Pods	54–159 min	\$131.73
Total	**Current Pod spend**	—	**\$237.61**

The refresh supersedes earlier closeout snapshots whose last hourly buckets had not all posted, including values previously described as settled. It found zero Pods and one intentionally retained 100 GB Standard volume. Run 018 accounts for 15 H200 allocations: one timing preflight, four sentinel Pods, eight planned remainder Pods, and two infrastructure retries. Report-local `billing-refresh.json` retains the query window, Pod IDs, exact GPU/disk amounts, and reconciliation totals.

4 Results

The analysis adds one intervention at each stage and maps the resulting validation-loss–logical-compute frontier. Only completed comparisons appear as result subsections.

All site columns report exact-zero percentages; `q` and `k` denote the post-RoPE tensors. "n.m." denotes a site that was not measured in the source run.

Table 5: Progressive analysis status.

Stage	**Comparison**	**Evidence status**
1	A0 → A1-H	Full-pass endpoints complete
Control	A0/A1-H → uniform post-hoc clipping	Full-pass Analysis 005 complete
2	A1-H → A1-H-L1	Full-pass dose response complete
3	A1-H-L1 → A1-H-OL1	Matched full-pass Analysis 003 complete
4	A1-H → A4	Matched full-pass Analysis 004 complete
5	A4 → A4-OL1	Corrected Run 015 in Analysis 009; descriptive; F001 discarded
6	A4 → A7	Matched full-pass Analysis 008; tentative F002
7	A7 → A7-OL1	Five matched full-pass pairs in Analysis 008; descriptive
Scale	14M → selected 70M ladder	Four-family promotion complete in Analysis 010; descriptive

4.1 Type of non-linearity: A0 to A1-H

ReLU creates a 63.448% exact-zero rate at `h` and moves R_{model} to 2.714%.

Table 6: Pythia-14M full-pass controls from Run 004.

Condition	**Loss**	R_{model} **(%)**	**Exact-zero (%)**						
			`h`	`m`	`a`	`z`	`q`	`k`	`v`
A0: GeLU	5.209	< 0.001	< 0.001	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
A1-H: ReLU	5.270	2.714	63.448	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001

The evaluation-only uniform TEAL-style control thresholds `a`, `m`, `h`, and `z` separately in every block; it does not threshold `q`, `k`, or `v`. For one common target p, values satisfying $|x| \le t$ are zeroed without retraining.

Table 7 is the only result table that reports paired loss deltas.

Table 7: Selected uniform TEAL-style post-hoc endpoints from Analysis 005, Observation O001. Deltas are paired to the independently evaluated $p = 0$ row for the same checkpoint.

	GeLU control			ReLU control		
$\boldsymbol{p}$	**Loss**	**Δloss**	R_{model} (%)	**Loss**	**Δloss**	R_{model} (%)
0.0	5.208594	0.000000	< 0.001	5.269633	0.000000	2.714133
0.1	5.214453	0.005859	1.278706	5.274073	0.004439	3.568403
0.2	5.250270	0.041676	2.554786	5.307940	0.038307	4.418365
0.3	5.361901	0.153307	3.831641	5.405056	0.135422	5.271871
0.4	5.605384	0.396790	5.125544	5.602965	0.333331	6.129325
0.5	6.077731	0.869137	6.441923	5.976274	0.706641	6.994451

4.2 Naive L1 pressure: A1-H to A1-H-L1

Increasing λ raises exact-zero mass and R_{model} across the four full-pass endpoints.

Table 8: Run 004 full-pass naive-L1 dose response.

$\boldsymbol{\lambda}$	**Loss**	R_{model} (%)	Exact-zero (%)						
			`h`	`m`	`a`	`z`	`q`	`k`	`v`
0.05	5.206	3.142	73.441	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
0.10	5.165	3.336	77.993	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
0.50	5.113	3.788	88.542	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
1.00	5.102	3.949	92.323	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001

4.3 Orthogonalization: A1-H-L1 to A1-H-OL1

The matched full-pass endpoints occupy a narrow common region in validation loss, R_{model}, and `h` sparsity.

Table 9: Absolute naive-L1 and OL1 endpoints from Analysis 003.

$\boldsymbol{\lambda}$	**Method**	**Loss**	R_{model} (%)	Exact-zero (%)						
				`h`	`m`	`a`	`z`	`q`	`k`	`v`
0.05	L1	5.206	3.142	73.441	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
	OL1	5.198	3.145	73.526	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
0.10	L1	5.165	3.336	77.993	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
	OL1	5.159	3.339	78.046	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
0.50	L1	5.113	3.788	88.542	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
	OL1	5.110	3.768	88.084	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
1.00	L1	5.102	3.949	92.323	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001
	OL1	5.121	3.938	92.067	0.000	n.m.	n.m.	< 0.001	< 0.001	< 0.001

Figure 2 isolates the OL1 Adam-relative geometry before and after conflict projection.

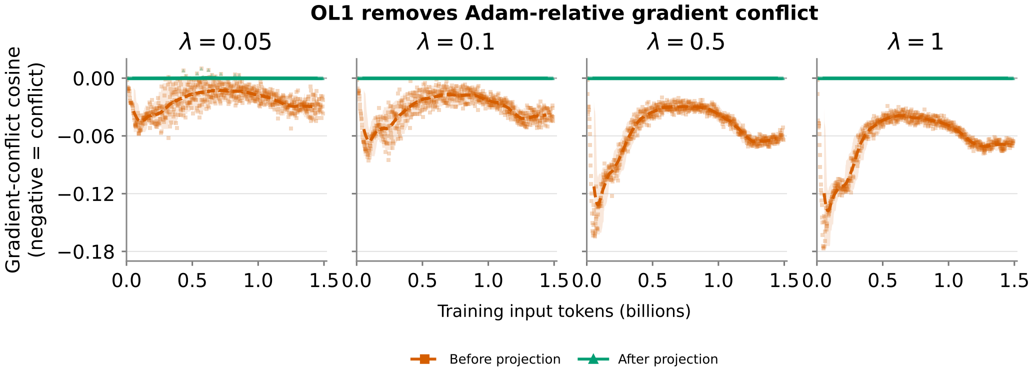

Figure 2: OL1 Adam-relative interaction before and after conflict projection. Each panel contains 712 optimizer boundaries for one λ; lines are centered 51-boundary means and bands span p05–p95. Source: Analysis 003, Observation O004.

4.4 Architecture-wide thresholding and OL1: A4 to A4-OL1

A4 applies one-sided thresholds at `a`, `m`, `h`, and `z`. The corrected A4-OL1 objective in Run 015 adds post-gate OL1 pressure at all four sites with $\lambda = 1$ and trust budget 1.0. It improves both validation loss and R_{model} over matched A4 at $\kappa = 0$ and 0.01. At $\kappa = 0.05$, 0.1, and 0.5, it adds 2.2506, 2.4413, and 2.4979 percentage points of R_{model} for validation-loss costs of 0.055693, 0.128681, and 0.378307. This one-seed result remains descriptive; discarded Finding F001 is not restored.

Table 10: Absolute A4 and corrected four-site A4-OL1 full-pass endpoints from Analysis 009, Observation O001. Every percentage is recomputed after pooling integer counts over all six layers and 338 complete validation blocks.

κ	Method	Loss	R_{model} (%)	Exact-zero (%)						
				`h`	`m`	`a`	`z`	`q`	`k`	`v`
0.00	A4	5.470	7.212	68.765	49.779	48.560	54.525	< 0.001	< 0.001	< 0.001
	A4-OL1	5.458	7.810	74.205	39.296	68.891	69.625	< 0.001	< 0.001	< 0.001
0.01	A4	5.466	7.414	71.181	50.397	49.095	59.643	< 0.001	< 0.001	< 0.001
	A4-OL1	5.458	8.587	80.833	43.464	73.491	85.259	< 0.001	< 0.001	< 0.001
0.05	A4	5.434	8.206	81.340	52.712	51.155	77.641	< 0.001	< 0.001	< 0.001
	A4-OL1	5.490	10.456	93.275	60.439	88.505	97.392	< 0.001	< 0.001	< 0.001
0.10	A4	5.420	8.953	91.853	55.200	53.244	89.234	< 0.001	< 0.001	< 0.001
	A4-OL1	5.548	11.394	97.149	75.912	91.310	99.283	< 0.001	< 0.001	< 0.001
0.50	A4	5.660	10.216	99.876	66.386	63.434	99.881	< 0.001	< 0.001	< 0.001
	A4-OL1	6.038	12.713	99.940	96.766	98.906	99.976	< 0.001	< 0.001	0.661

4.5 Attention-operand thresholding and OL1: A4 to A7 to A7-OL1

A7 adds symmetric thresholds at post-RoPE `q`, `k`, and `v` to A4's one-sided `a`, `m`, `h`, and `z` gates; the matched A4/A7 result supports tentative Finding F002. A7-OL1 adds post-gate orthogonal L1 pressure at all seven active sites with $\lambda = 1$ and trust budget 1. At matched $\kappa = 0.1$, A7-OL1 adds 1.3717 percentage points of R_{model} for +0.000816 validation loss; at $\kappa = 0.5$, it adds 12.0959 points for +0.126466 loss. The zero-threshold row regresses, so the A7/A7-OL1 effect is threshold-dependent and remains descriptive.

Table 11: Matched A7 and A7-OL1 full-pass endpoints from Analysis 008, Observation O001. Every percentage is recomputed after pooling integer counts over all six layers and 338 complete validation blocks. `out` is the ungated post-W_o `attention_output`.

κ	Variant	Loss	R_{model} (%)	Exact-zero mass (%)							
				`h`	`m`	`a`	`z`	`q`	`k`	`v`	`out`
0.00	A7	5.468401	7.217651	68.689	49.790	48.590	55.221	< 0.001	< 0.001	< 0.001	< 0.001
	A7-OL1	5.480184	7.054229	68.277	49.503	48.583	42.761	< 0.001	< 0.001	< 0.001	< 0.001
0.01	A7	5.458822	7.617649	71.275	50.330	49.064	59.041	0.399	0.407	1.654	< 0.001
	A7-OL1	5.475797	7.723393	71.050	50.103	49.196	48.933	1.373	1.496	2.338	< 0.001
0.05	A7	5.437888	9.126924	81.298	52.681	51.036	78.018	1.840	1.716	7.312	< 0.001
	A7-OL1	5.462811	9.863032	81.052	52.241	51.036	72.067	5.427	5.817	10.526	< 0.001
0.10	A7	5.428681	10.425018	91.353	55.075	53.052	90.052	3.150	3.227	11.355	< 0.001
	A7-OL1	5.429497	11.796760	91.607	54.874	53.104	88.896	8.227	8.544	18.819	< 0.001
0.50	A7	5.702923	15.386813	99.859	65.721	62.306	99.865	17.435	18.134	31.255	< 0.001
	A7-OL1	5.829390	27.482684	99.876	68.698	75.732	99.918	93.545	94.541	98.713	< 0.001

4.6 Selected scale promotion: Pythia-14M to 70M

Run 018 promotes A0, A1-H, A4-OL1, and A7-OL1 to randomly initialized Pythia-70M under the matched one-pass contract. Analysis 010 compares those endpoints with their Pythia-14M counterparts. This is a selected ladder promotion, not a complete 70M rerun of every intermediate ablation and not a scaling-law estimate.

The qualitative crossover persists. A4-OL1 dominates A7-OL1 at $\kappa = 0$ at both sizes, while A7-OL1 dominates A4-OL1 at $\kappa = 0.5$. Within A7-OL1, the $\kappa = 0.1$ point gains 4.743 percentage points of R_{model} over $\kappa = 0$ while lowering 14M loss by 0.05069; at 70M it gains 5.131 points for only +0.00890 loss. Across $\kappa = 0$ to 0.5, A4-OL1 gains 4.903 and 9.924 points of R_{model} at 14M and 70M, while A7-OL1 gains 20.428 and 17.039 points. The result is descriptive one-seed evidence across two sizes; 410M remains unobserved.

Table 12: Pythia-14M and 70M A4-OL1 full-pass endpoints from Analysis 010, Observation O001. Percentages pool integer counts over all six layers and 338 complete validation blocks.

κ	Scale	Loss	R_{model} (%)	Exact-zero mass (%)						
				`h`	`m`	`a`	`z`	`q`	`k`	`v`
0.00	14M	5.458170	7.810042	74.205	39.296	68.891	69.625	< 0.001	< 0.001	< 0.001
	70M	4.805289	25.672509	92.586	49.378	56.605	93.390	< 0.001	< 0.001	< 0.001
0.01	14M	5.458277	8.586709	80.833	43.464	73.491	85.259	< 0.001	< 0.001	< 0.001
	70M	4.822732	26.224248	94.129	50.779	57.965	95.418	< 0.001	< 0.001	< 0.001
0.05	14M	5.489803	10.456445	93.275	60.439	88.505	97.392	< 0.001	< 0.001	< 0.001
	70M	4.873932	28.489958	97.726	56.154	69.631	97.935	< 0.001	< 0.001	< 0.001
0.10	14M	5.548323	11.394281	97.149	75.912	91.310	99.283	< 0.001	< 0.001	< 0.001
	70M	4.943729	30.574258	98.933	64.998	78.393	98.913	< 0.001	< 0.001	< 0.001
0.50	14M	6.037987	12.713449	99.940	96.766	98.906	99.976	< 0.001	< 0.001	0.661
	70M	5.389543	35.596219	99.966	91.696	95.184	99.937	< 0.001	< 0.001	0.181

Table 13: Pythia-14M and 70M A7-OL1 full-pass endpoints from Analysis 010, Observation O001. Percentages pool integer counts over all six layers and 338 complete validation blocks.

κ	Scale	Loss	R_{model} (%)	Exact-zero mass (%)						
				`h`	`m`	`a`	`z`	`q`	`k`	`v`
0.00	14M	5.480184	7.054229	68.277	49.503	48.583	42.761	< 0.001	< 0.001	< 0.001
	70M	4.941204	23.562417	85.969	50.300	49.731	68.611	< 0.001	< 0.001	< 0.001
0.01	14M	5.475797	7.723393	71.050	50.103	49.196	48.933	1.373	1.496	2.338
	70M	4.936382	24.226276	86.643	50.915	50.314	72.792	1.504	1.684	2.063
0.05	14M	5.462811	9.863032	81.052	52.241	51.036	72.067	5.427	5.817	10.526
	70M	4.963642	26.523941	89.474	52.992	52.340	80.939	7.901	8.684	10.135
0.10	14M	5.429497	11.796760	91.607	54.874	53.104	88.896	8.227	8.544	18.819
	70M	4.950103	28.693785	92.261	55.165	54.309	88.075	14.074	14.568	19.053
0.50	14M	5.829390	27.482684	99.876	68.698	75.732	99.918	93.545	94.541	98.713
	70M	5.215976	40.601872	99.865	69.011	67.243	99.845	71.531	72.123	86.939

The 70M post-hoc controls preserve the same local shape as 14M: low target sparsity first increases logical opportunity at small loss cost, followed by a steep quality penalty. TEAL clips `a,m,h,z` at evaluation only; it is not trained gating. Its source artifacts did not record downstream `q,k,v` activation mass, so those quantities are not inferred from R_{model}.

Table 14: Complete Pythia-70M uniform TEAL-style post-hoc frontiers from Analysis 010. Deltas are paired to the independently evaluated $p = 0$ row for the same checkpoint.

	A0 / GeLU			A1-H / ReLU		
p	Loss	Δloss	R_{model} (%)	Loss	Δloss	R_{model} (%)
0.0	4.099766	+0.000000	< 0.001	4.222750	+0.000000	10.065758
0.1	4.103608	+0.003842	3.704772	4.223747	+0.000998	12.536992
0.2	4.130721	+0.030955	7.409856	4.234510	+0.011760	15.009890
0.3	4.207335	+0.107569	11.128199	4.269464	+0.046714	17.485463
0.4	4.397795	+0.298029	14.844233	4.346144	+0.123394	19.985072
0.5	4.833057	+0.733291	18.588525	4.522054	+0.299304	22.515643
0.6	5.657785	+1.558019	22.410886	4.914340	+0.691591	25.107710
0.7	6.927989	+2.828223	26.348400	5.722641	+1.499891	27.893498
0.8	8.072236	+3.972470	30.462129	6.912035	+2.689285	31.035075
0.9	9.179397	+5.079631	34.581403	9.133301	+4.910551	33.891163

4.7 Trained and post-hoc full-pass frontier

Figure 3 places both model sizes on one absolute validation-loss–R_{model} plane. It includes all 20 trained A4-OL1/A7-OL1 endpoints and all 40 A0/A1-H post-hoc TEAL points. The vertical display ends at loss 6, but the full trajectories are retained and visibly exit the panel; every coordinate remains in Analysis 010's machine-readable reduction and complete Markdown tables.

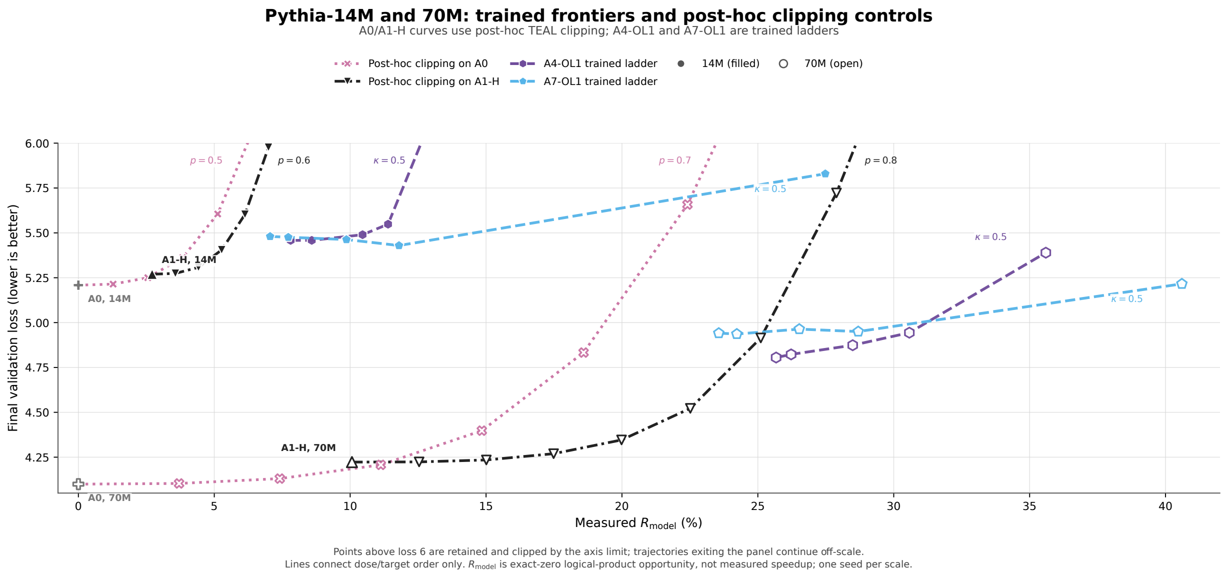

Figure 3: Selected Pythia-14M and 70M trained and post-hoc frontier. Filled markers identify 14M and open markers 70M; magenta/black trajectories are A0/A1-H post-hoc TEAL and purple/sky-blue trajectories are trained A4-OL1/A7-OL1. Points above loss 6 remain in their plotted series and continue off-scale. Lines indicate dose or target order only. R_{model} is logical zero-product opportunity, not measured speedup. Source: Analysis 010, Observation O001.

A Historical A4-OL1[h] realization (Run 012)

Run 012 declared A4-OL1 pressure at `a,m,h,z`, but its frozen training code constructed the pressure capture with `["h"]`; the realized intervention was therefore A4 gates plus OL1 pressure only at `h`. The five endpoints remain valid for that realized A4-OL1[h] intervention, but they are excluded from the main four-site A4-OL1 result and cannot support discarded Finding F001.

Table 15: Historical Run 012 A4-OL1[h] full-pass endpoints. Exact-zero percentages pool integer counts over all six layers and 338 complete validation blocks. Source: Analysis 009, Observation O001.

κ	Loss	R_{model} (%)	Exact-zero (%)						
			`h`	`m`	`a`	`z`	`q`	`k`	`v`
0.00	5.216	8.409	93.543	49.816	49.413	64.671	< 0.001	< 0.001	< 0.001
0.01	5.205	8.586	94.871	50.390	49.942	71.971	< 0.001	< 0.001	< 0.001
0.05	5.196	9.036	97.915	52.391	51.791	88.307	< 0.001	< 0.001	< 0.001
0.10	5.229	9.343	99.125	54.747	53.809	96.774	< 0.001	< 0.001	< 0.001
0.50	5.723	10.227	99.944	66.441	63.622	99.934	< 0.001	< 0.001	< 0.001

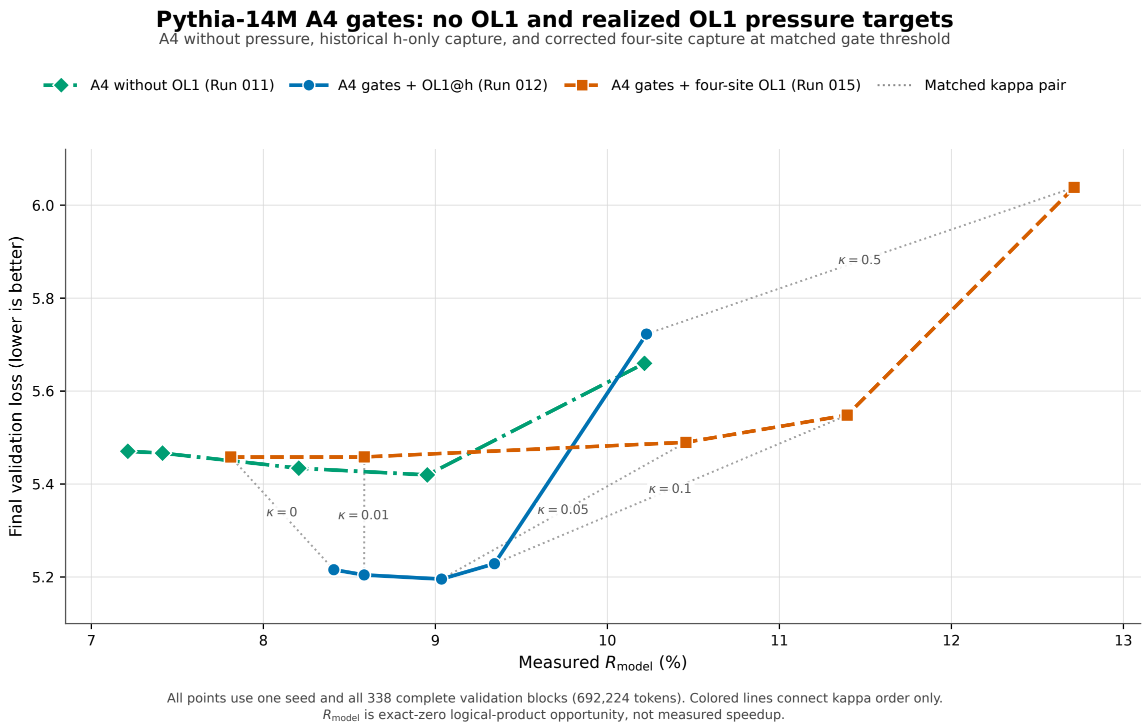

Figure 4: Matched A4, historical Run 012 A4-OL1[h], and corrected Run 015 four-site A4-OL1. The historical and corrected pressure objectives are scientifically different interventions. Source: Analysis 009, Observation O001.

Run 012 used five A100 80 GB Pods for 70–100 minutes per condition and cost $14.85 in the current Pod-ID-reconciled billing refresh. Including this historical run, current Pod spend across all report rows is $252.46; the independent retained network-volume charge remains excluded.